



Semester I Examinations 2019/2020

Exam Code(s)	1SD1/1SD3/1MF1
Exam(s)	Higher Diploma in Applied Science (Software Design & Development), Higher Diploma in Applied Science (Software Design & Development) Industry Stream, M.Sc. in Software Design & Development
Module Code(s)	CT874
Module(s)	Programming I
Paper No.	I
External Examiner(s)	Prof. Pier Luca Lanzi
Internal Examiner(s)	Prof. Michael Madden * Dr. Séamus Hill
<u>Instructions:</u>	Candidates are required to answer: Section A (20 Marks) Answer all 10 parts of section A, This section consists of 10 multiple-choice questions all of which should be attempted. Answers are to be written on the MCQ sheet provided, NOT on the Examination paper. Section B (80 Marks) Answer Any Two Questions.
Duration	2 hours
No. of Pages	6
Discipline(s)	Computer Science
Requirements	MCQ

Section A (20 Marks)
Attempt all 10 questions
Mark your answers on the MCQ sheet provided

- 1) Which of the following is the correct syntax for creating an ArrayList?
 - a) `ArrayList<Person> list = new ArrayList<>();`
 - b) `ArrayList<> list = new ArrayList<>();`
 - c) `ArrayList<> list = new List<>();`
 - d) `ArrayList<> list = new ArrayList<Person>();`

- 2) Which of the following is the correct syntax for creating an array.
 - a) `Int[] myArray = new int[10];`
 - b) `[]int myArray = new int[10];`
 - c) `int[] myarray = new [10]int;`
 - d) `int[] myArray = int[10];`

- 3) How would you access the 10th element of the Array, a.
 - a) `a(10);`
 - b) `a[10];`
 - c) `a[10];`
 - d) `a(10);`

- 4) Assume we have three classes, X Y and Z. Y and Z are siblings and are subclasses of the X class. Which classes have visibility of a private variable in class X?
 - a) X, Y and Z
 - b) X and Y only
 - c) Y and Z only
 - d) X only

- 5) Which of the following methods is not contained within the String class?
 - a) `charAt(int index)`
 - b) `concat(String str)`
 - c) `length()`
 - d) `equal(int index)`

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- 6) The practice of writing multiple constructors to handle different sets of inputs is known as _____.
- a) Inheritance
 - b) Overriding
 - c) Overloading
 - d) Multiplicity
- 7) When an array is passed to a method _____.
- a) only the reference is passed, and a copy of the array is not created in the method.
 - b) a copy of the array is created within the method.
 - c) the value of the array is passed into the method.
 - d) None of the above.
- 8) What is meant by “returning an object” from a method?
- a) It means you are returning a null value to the caller.
 - b) It means returning a primitive type to the caller.
 - c) It means returning the object’s address to the caller.
 - d) It means returning the object’s value to the caller.
- 9) The use of “this()” _____.
- a) is a reference to hello
 - b) is bad style
 - c) is illegal
 - d) is a call to a Function constructor
- 10) Which of the following is not a Boolean operator?
- a) &&
 - b) !
 - c) ||
 - d) ++

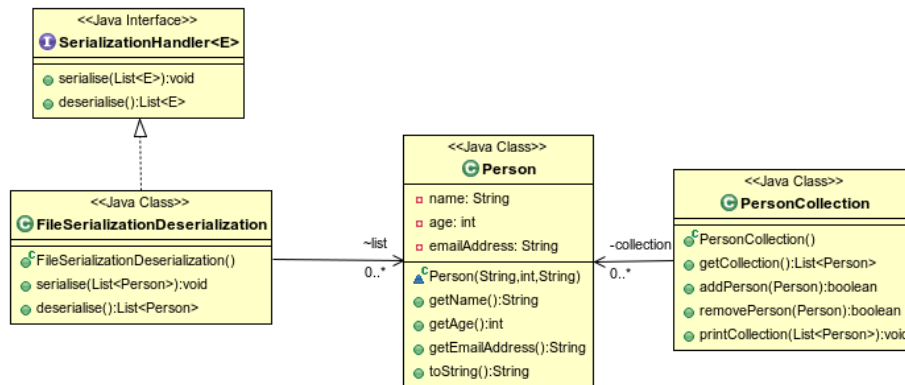
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Section B (80 Marks)

Answer Any TWO Questions

Q 2.

a) You are required to implement the interface and classes outlined below.



- i. The **FileSerializationDeserialization** class provides an implementation of the abstract `serialise()` and `deserialise()` methods found in the **SerializationHandler** Interface.
- ii. The **PersonCollection** class creates an `ArrayList` designed to contain objects of type `Person`.

[30]

b) Write a tester class which:

- i. creates an instance of the **PersonCollection** class and populate the `ArrayList` with 4 `Person` objects, displaying the List's contents.
- ii. using Composition Serialize the list.
- iii. Using Composition deserialize the list displaying it's contents.

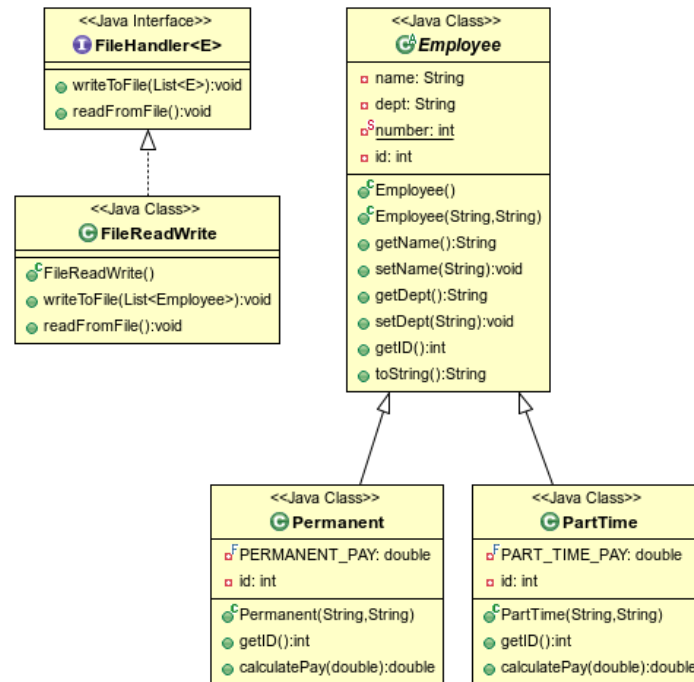
[10]

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Q 3.

From the UML diagram below you are required to:

a) Implement the Interface and Classes outlined below.



Note: the Employee class contains an abstract method ***calculatePay()***. Permanent employees have a fixed pay rate of €12.50 per hour and part time employees have an hourly rate of €10.50. The ***FileReadWrite*** class provides implementations that allow Lists to be written to a text file and read from a text file.

[25 Marks]

b) Write a driver class to carry out the following:

- Create an *LinkedList* and populate the list with two *Permanent* objects and two *PartTime* objects, displaying the contents of the list.
- Calculate and display the pay for each employee for a 35 hour week.
- Using *Composition* write the list to a text file and read it back displaying the contents.

[15 Marks]

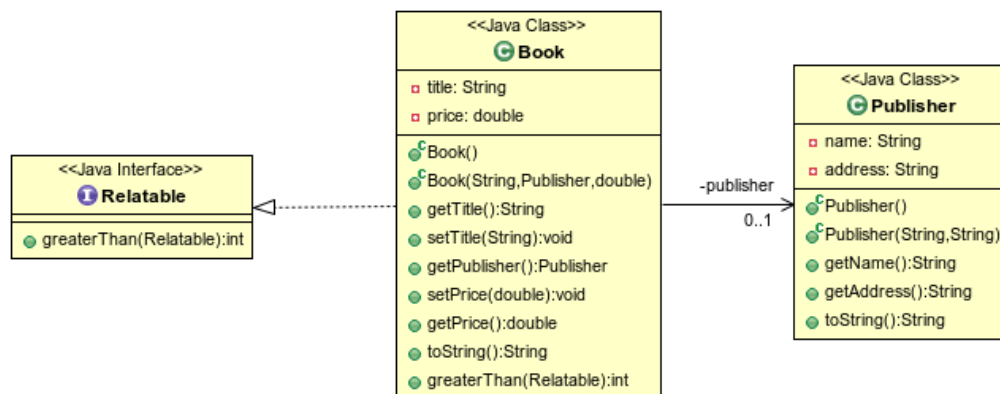
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Q 4.

a) In relation to Java describe in your own words, what an *Interface* is.

[4]

b) Using the UML diagram outlined, write a Java application that implements the **Book** class, **Publisher** class and **Relatable** interface to allow comparison between *Book* objects based on *price*.



Note: *greaterThan()* returns a 1, 0 or -1 value depending on whether an object is greater than, equal to or less than a similar type of object.

[18]

c) Write a driver class which:

i. Creates and populates an *ArrayList* with three instances of the *Book* class.

[6]

ii. Includes a method with the following signature:

public static *Book* findMostExpensive(List<*Book*> b)

The method uses a *for loop* to traverses the List calling **greaterThan** to determine the most expensive in the List, which is then returned.

[10]

iii. Displays the most expensive book in the List using *toString()*.

[2]